

PROVOCATION POLIOMYELITIS: NEGLECTED CLINICAL OBSERVATIONS FROM 1914 TO 1950

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 The association between paralytic poliomyelitis in a recently injected limb and the injection as the precipitating factor, was first accepted in 1950 and rapidly confirmed by research.¹ Once it was recognized that injection of diphtheria-pertussis-tetanus vaccine (DPT) provoked poliomyelitis, incidents previously thought fortuitous were recognized as similar. Thus, the cases following the use of the Kolmer vaccine in 1936,² and many cases following treatment with neoarsphenamine for yaws in Samoa in 1932,³ were seen as provoked, as well as the cases reported following tonsillectomy.⁴

The immediate tasks for medical researchers when this association was noticed were to discover its extent and to reproduce it in experimental animals. If the association was proved, then vaccination with DPT would have to be modified. In this paper, I propose to examine the literature before 1952 when the term provocation was, I believe, first used for poliomyelitis, to see which clues had already been found indicating an association and which could have been recognized. This, I hope, will shed light on the question of why the significance of clinical and research observations has not always been immediately appreciated and the effect of this lack of recognition on the history of the disease.

Early Accounts. The earliest accounts of provocation poliomyelitis have, I believe, been overlooked until now. In May 1914, Dr. Hans Kern reported an epidemic of poliomyelitis in a children's institution of twenty-two beds in Germany.⁵ Between 22 September and 10 October 1913, five cases of paralytic poliomyelitis were diagnosed in children between one and three years of age. All the children in the home were suffering from congenital syphilis and

¹ J. K. Martin, "Local paralysis in children after injections," *Arch. Dis. Child.*, 1950, 25: 1-14; D. H. Geffen, "The incidence of paralysis occurring in London children within four weeks after immunization," *Med. Officer*, 1950, 83: 137-40; B. P. McCloskey, "The relation of prophylactic inoculations to the onset of poliomyelitis," *Lancet*, 1950, 1: 659.

² N. Nathanson and A. D. Langmuir, "The Cutter incident: poliomyelitis following formaldehyde-inactivated poliovirus vaccination in the United States during the spring of 1955. I. Background," *Amer. J. Hyg.*, 1963, 78: 16-28.

³ S. M. Lambert, "A yaws campaign and an epidemic of poliomyelitis in Western Samoa," *J. Trop. Med. Hyg.*, 1936, 39: 41-46.

⁴ W. L. Aycock and E. H. Luther, "The occurrence of poliomyelitis following tonsillectomy," *New Eng. J. Med.*, 1929, 200: 164-67.

⁵ H. Kern, "Ueber eine anstaltsendemie von Heine-Medizinischer krankheit," *Muen. Med. Wochens.*, 1914, 61: 1053-55.

were being treated with Neosalvarsan, Salvarsan and other drugs. All five cases had been treated with Neosalvarsan, cases 3 and 4 intravenously, and 1, 2, and 5 subcutaneously. In his report, Kern noted "the local way in which Neosalvarsan could give rise to paralysis of the leg." The period from injection of Neosalvarsan to illness was from five to twenty-one days and from six to thirty-one days to paralysis in cases 1, 3, 4 and 5, and in case 2, it was two and fourteen days respectively, after an injection of Hg-salicyl-Kur.

In 1928, in another report from Germany, Alterthum gave details of seven cases of poliomyelitis in children with congenital syphilis having been treated with Neosalvarsan and other drugs between 1921 and 1926.⁶ He noted the "striking temporal relationship between treatment of syphilis and the appearance of poliomyelitis" and that "3 children (cases 1, 2, 8) fell ill during treatment and in 3 further (cases 4, 5, 7) the conclusions of the course lay a short time (2, 4, 6 weeks) before poliomyelitis." In 1936, H. Gougerot reported three cases of poliomyelitis following arsenical therapy of syphilis.⁷ S. J. Wilkinson recorded a single case of paralysis following alum-precipitated diphtheria and pertussis vaccine in 1937.⁸ G. S. Wilson, in his study of the hazards of immunization, considers this "very suggestive" of provocation poliomyelitis, but as the child recovered completely within two to three months, it would not now be accepted as poliomyelitis.⁹

The observations of these researchers were very acute, for the characteristics of provocation poliomyelitis are: 1. first symptoms of paralysis seven–eighteen days after the provoking injections, except where the provoking injection itself contained live poliovirus; 2. a high incidence of initial paralysis in the limb given the provoking injection; 3. a higher rate of paralysis among those given provoking injections; 4. a greater proportion of limbs with residual paralysis in those given provoking injections. An explanation of provocation has recently been given.¹⁰ Why were these early important observations unheeded?

It was unfortunate that the papers of both Kern and Altherthum were published in German, without abstracts and with inadequate titles. Moreover, the papers were difficult to read, and had very poor bibliographies. It is probable that Kern's paper was forgotten in the upheaval in Europe in World War I. Gougerot's paper, without those of Kern and Altherthum to support it, appeared anecdotal. Similarly, observations in the 1950s on the susceptibility to poliomyelitis of children with congenital syphilis by V. Menichella and D. Nabarro were ignored.¹¹ All these authors were clinical

⁶ Alterthum, "Lues congenita und poliomyelitis," *Deut. Med. Wochen.*, 1928, 54: 522–23.

⁷ H. Gougerot, "Eveil d'infection neurotrope à virus filtrant à la suite d'arsenothérapie chez des syphilitiques," *Bull. Soc. Derm. Syph.*, 1935, 42: 794–95.

⁸ S. J. Wilkinson, "Multiple neuritis from diphtheria toxoid: report of a case," *J. Pediat.*, 1937, 10: 180–83.

⁹ G. S. Wilson, *The Hazards of Immunization* (London: Athlone Press, 1967), pp. 270–78.

¹⁰ H. V. Wyatt, "Provocation poliomyelitis and entry of poliovirus to the CNS," *Med. Hypotheses*, 1976, 2: 269–74.

¹¹ V. Menichella, "Epidemiology of poliomyelitis in a children's community," in *Poliomyelitis: Papers and Discussions Presented at the Third International Poliomyelitis Conference* (Philadelphia: Lippincott, 1955), p. 551; D. Nabarro, *Congenital Syphilis* (London: Arnold, 1954).

veneriologists working outside the mainstream of poliomyelitis research, either laboratory or epidemiological. Although the number of cases was small, the observations were accurate and acute. Moreover, the case-rate was extremely high; in Kern's epidemic, it was twenty-three percent of children under four years.¹² Menichella presented his data as an exhibit at the International Poliomyelitis Congress held in Rome in 1954, showing an increased rate in syphilitic children of $\times 23$ compared with normal. That syphilitic children were treated with injections would be known to medical scientists who nevertheless failed to connect this with provocation poliomyelitis, although the original work on provocation published in 1950 had been confirmed.

The Bibliography of Infantile Paralysis. The recognition of provocation poliomyelitis coincided with the publication of the massive second edition of this bibliography.¹³ Published in 1951, it listed 10,367 papers up to the end of 1949, thereby missing the actual papers on provocation. Nevertheless, the concept was important and already established, and should have been known to the editors or at least communicated to them. Martin's paper (n. 1) appeared in the March issue, 1950, of *Archives of Diseases of Children* and Geffen's and McCloskey's papers (n. 1) both appeared in weekly journals on 8 April 1950.¹⁴ Geffen's paper had been read at a meeting of medical officers on 7 October 1949 and McCloskey's had been submitted to the medical authorities in Australia on 20 July 1949 and discussed by them with Professor F. M. Burnet in September 1949. Moreover, a letter had been sent to all doctors in Victoria and a press statement had been made on 4 October 1949. Although all three papers were published in British journals, these must have been available in the U.S.A. shortly afterwards. Geffen says that "correspondence with experts on poliomyelitis in America tends to show that so far no correlation between inoculation and the paralysis of poliomyelitis has been observed."¹⁵ A major paper appeared in the *British Medical Journal* on 1 July 1950 in which the authors wrote "In the autumn of 1949 . . . it was known that much more incriminating evidence was accumulating in several quarters." After detailing the evidence of Martin, Geffen and McCloskey "which has since been published" the authors proceeded "These reports, at that time incomplete and unpublished, made it essential that a wide-scale statistical inquiry be immediately undertaken."¹⁶

The term "provocation" was not used by Martin, Geffen and McCloskey. It was perhaps unfortunate that "poliomyelitis" was not included in the title of two of the papers and Geffen's paper was not even indexed under poliomyelitis in *Medical Officer!* Martin discussed neurological complications following the administration of serum and other inoculations and the

¹² Kern, "Anstaltendemie," pp. 1053–55.

¹³ M. Fishbein, E. M. Salmonsens, and L. Hektoen, eds., *A Bibliography of Infantile Paralysis, 1789–1949*, 2nd ed. (Philadelphia: Lippincott, 1951).

¹⁴ See note 1.

¹⁵ Geffen, "Incidence of paralysis," p. 140.

¹⁶ A. B. Hill and J. Knowelden, "Inoculation and poliomyelitis: a statistical investigation in England and Wales in 1949," *Brit. Med. J.* 1950, 2: 1–6.

analogy of the relationship of tonsillectomy and poliomyelitis. Geffen did not discuss the pathology, but McCloskey discussed exercise, trauma and bruising.

The first use of the terms "provoking" or "provocation" that I can find was at the Second International Poliomyelitis Conference held in 1951, whose proceedings were published in 1952, when it was used by Bradford Hill in the title of his talk and by two speakers in the discussion.¹⁷ The term was not used in papers published earlier in 1952¹⁸ and is, interestingly, absent in the index of J. R. Paul's magnificent history of poliomyelitis.¹⁹ Provocation, therefore, did not appear in the subject index of the *Bibliography of Infantile Paralysis*. However, the subject index occupied 120 pages and relevant papers might have been listed and found under other headings. The following entries were most likely: *Predisposition* (see also Heredity and Susceptibility): there was one paper (1945) reviewing heredity, pregnancy, tonsillectomy, trauma, etc., and *Susceptibility*: there were twenty-seven subheadings but only one paper bearing on provocation—the effect of physical exercise (1938).

There was a considerable section under *Tonsillectomy*, although the first paper to connect tonsillectomy with poliomyelitis by M. Talmey was not indexed under tonsillectomy:²⁰ it was recorded in Paul's history.²¹ *Immunization* was devoted almost entirely to poliovirus immunization in man and animals, the only possible subheading concerned an increase in susceptibility in *monkeys* following typhoid-paratyphoid immunization. Gougerot's paper was the only one listed under *Arsenical therapy*: the paper by Kern was not listed under *Syphilis*, but only under *Epidemics*. There were sixteen papers listed under *Syphilis* but almost all described single cases, and were anecdotal. A further paper was listed under *Tabes*. There were no headings for Neosalvarsan or Salvarsan. Lambert's paper was the only entry under *Yaws*. There were many entries under *Traumatic Poliomyelitis*, but these referred mainly to anecdotal papers of individual coincidence of physical trauma and paralysis.

The papers by Kern and Alterthum may be found in the *Bibliography* according to their year of publication, and the entries give the translated titles. Gougerot's paper, given a seven-line abstract, was incorrect as one patient with no case number was missed by the abstracter. I first found Kern's paper in the *Bibliography* because I was looking for data about epidemics in institutions. Such data can be analyzed to test the model of genetic susceptibility to poliomyelitis.²² It was traced via the subject index

¹⁷ A. B. Hill, "Inoculation procedures as a provoking factor in poliomyelitis," in *Poliomyelitis: Papers and Discussions Presented at the Second International Poliomyelitis Conference* (Philadelphia: Lippincott, 1952), pp. 330–32, Discussion, pp. 333–40.

¹⁸ For example, M. Greenberg, H. Abramson, H. M. Cooper, and H. E. Solomon, "The relation between recent injections and paralytic poliomyelitis in children," *Amer. J. Pub. Health*, 1952, 42: 142–52.

¹⁹ J. R. Paul, *A History of Poliomyelitis* (New Haven, Conn.: Yale University Press, 1971).

²⁰ M. Talmey, "Predisposing factors in infantile paralysis," *New York Med. J.*, 1916, 104: 202–4.

²¹ Paul, *History of Poliomyelitis*, p. 188.

²² H. V. Wyatt, "Is poliomyelitis a genetically-determined disease: I. A genetic model and II. A critical examination of the epidemiological data," *Med. Hypotheses*, 1975, 1: 35–42, 87–96.

under *Epidemics in institutions*. I had read Menichella's paper at an earlier date without realizing its significance. The very high case rate reported in the papers of Kern, Alterthum and Menichella did not fit the predicted rates for young children, although Kern's case rate of twenty-three percent was very close to the predicted rate for adults comprising two percent homozygotes and twenty-four percent heterozygotes.²³ However, the pathogenesis of poliomyelitis proposed in later papers suggested that there might be a causal connection between congenital syphilis and high rates of poliomyelitis following injections.²⁴ The other references were found by searching the *Bibliography* under *Syphilis* and *Tabes*. A fuller account of the relationship of syphilis and poliomyelitis will need to be developed.

Injections for Yaws. S. Lambert's paper described a campaign to treat yaws in Samoa with injections, preferably two or more, or neoarsphenamine. In 1932, two or more injections in the buttock were given to nearly 36,000 persons.

We had the misfortune to have coincident with our work, an epidemic of poliomyelitis which lasted for several weeks. This broke out first in the areas in which we had begun to work, but the rumor did not spread quickly . . . In 1933, however, a great deal of our work was nullified by the mistaken idea on the part of the natives that the paralysis was due to our yaws injections.

The first case of paralysis occurred a week after the second yaws injection . . . At first the paralysis was attributed either to yaws treatment or to the method of injection, since all the patients first seen had paralysis in the lower limbs and all had had double buttock injections . . . In 37 villages in which injections were carried out, paralysis occurred. A survey shows no record of paralysis in other areas. . . . There were some cases of paralysis among persons who did not receive injections; some of those paralyzed received only one injection. During the 1932 campaign scarcely a child from a few months of age to adolescence was not injected and so all illnesses succeeding the campaigns were debited to injections. It is obvious, however, that the paralyses and injections cannot be related.²⁵

Thus Lambert gave a very convincing account of provocation poliomyelitis in 138 persons with paralysis out of 1,766 children under five years of age—and rejected any connection! Kern with only five cases and Alterthum with seven cases recognized the possible connection and discussed it. If only Lambert had been familiar with the papers of Kern and Alterthum, he might have realized the connection and provocation might have been recognized fourteen years earlier with the saving of many lives and cases of paralysis.

However, his recording of the outbreak may have influenced the acceptance of the Australian McCloskey's observations. Dr. Jean Macnamarra of Australia had contributed two reports to the International Committee for the Study of Infantile Paralysis which produced the 560-page compendium on

²³ *Ibid.*

²⁴ Wyatt, "Provocation poliomyelitis," pp. 269–74; H. V. Wyatt, "Is poliomyelitis an auto-allergic disease triggered by a virus?" *Med. Hypotheses*, 1976, 2: 262–68.

²⁵ Lambert, "Yaws campaign," pp. 44–45.

poliomyelitis which had such a powerful influence on later research. The longer item was a one-page report on an outbreak of poliomyelitis in the British Solomon Islands and New Guinea in 1929.²⁶ Dr. Macnamarra and Professor F. M. Burnet were engaged together in poliomyelitis research over many years and made some very important observations. Burnet described Dr. Macnamarra as one of "the two most distinguished women who ever went through an Australian medical course."²⁷ It is probable that with her interest in poliomyelitis in South East Asia, she would have known of Lambert's paper. McCloskey, at the end of his paper wrote, "Also I wish to thank Dame Jean Macnamarra (who is a constant inspiration to workers in the poliomyelitis field in Australia)."²⁸ Although McCloskey did not cite Lambert's paper, it is possible that Dr. Macnamarra could have been subconsciously receptive to the idea of provocation if she had had knowledge of Lambert's paper.

The epidemic in Samoa described by Lambert is, however, unique. In all other epidemics—except the special circumstances of the Cutter and Kolmer Incidents in the U.S.A. where poliomyelitis followed vaccination—it is believed that provocation occurred in persons who were already infected with poliovirus and had or were about to have a viremia.²⁹ In Samoa, the description quoted above shows that the epidemic followed the yaws campaign and was confined to the villages visited by the team. It seems therefore that the team was not only provoking poliomyelitis but must itself have carried the poliovirus from village to village. Carriage from place to place via a contaminated syringe is possible but unlikely: one or more of the team must have been a chronic excreter of poliovirus. The first experiments on provocation in primates by J. Melnick and N. Ledinko were negative.³⁰ Dorothy Horstmann described these experiments at the International Poliomyelitis Conference in 1951 in Rome and one discussant said "(Dr. Horstmann) has just told us that the virus most likely is brought into the organism by means of the needle and that perhaps, therefore, the route of injection might be without bearing on the paralysis." Another discussant, Dr. Kelleher, said that he had been observing over a period of ten years many cases in which his colleagues had "felt convinced of the relationship between deep intramuscular inoculations with diphtheria and diphtheria-pertussis antigen and the site of paralysis. We have been struck also with the frequency in which the paralysis has been extreme in the affected limb . . . it has been rare in our experience for paralysis to be found in areas remote from the site of inoculation."³¹

²⁶ International Committee for the Study of Infantile Paralysis, *Poliomyelitis* (Baltimore: Williams and Wilkins, 1932).

²⁷ F. M. Burnet, *Changing Patterns: An Atypical Autobiography* (London: Heinemann, 1968), p. 28.

²⁸ McCloskey, "Prophylactic inoculations," p. 659.

²⁹ Nathanson and Langmuir, "Cutter Incident, I."

³⁰ J. L. Melnick and N. Ledinko, "Vaccination as a provoking factor in poliomyelitis: an experimental approach," *J. Infect. Dis.*, 1952, 90: 279–82.

³¹ *Second International Poliomyelitis Conference*, pp. 333–40. Quote on Horstmann is on p. 336; Kelleher's comment on p. 339.

The first comprehensive search for earlier instances of provocation was by M. Greenberg and other researchers in 1952.³² In their paper they quoted eight previous reports, probably obtained from the *Bibliography of Infantile Paralysis*. However, most of the reports refer to smallpox vaccinations with coincident poliomyelitis and subsequent research has failed to show any evidence of provocation by smallpox vaccination. The earliest report quoted by Greenberg and his colleagues referred to acute poliomyelitis following an injection of anti-cholera vaccine in 1921, and probably correctly described provocation.³³ Horstmann quoted an outbreak of poliomyelitis following injections of arsphenamine in the treatment of yaws but did not give a citation.³⁴ Lambert is first cited by L. Rosen and G. Thooris in a paper published in 1953, but received for publication in October 1952.³⁵ However, these researchers did not cite Lambert in a previous 1952 publication.³⁶ Rosen and Thooris were describing an epidemic of poliomyelitis which occurred mainly on the island of Tahiti following a campaign of intramuscular injections of acetylarsen and other compounds against treponematoses. Their finding of Lambert's paper could have been through the coincidence of the two campaigns against treponemes with intramuscular injections of arsenic compounds in the Far East in island or isolated communities. It is also interesting to note that arsenicals and bismuth compounds, which were also probably provoking agents, were used on a large scale in other areas, Africa for instance, without any reported poliomyelitis.³⁷ In these areas poliomyelitis was probably endemic and a few cases of provocation poliomyelitis went unnoticed. Findlay had noticed occasions when poliomyelitis had followed injections of guanine in African children but it is not clear whether this was before or after the 1950 reports on provocation.³⁸

As this discussion shows, an attempt by researchers to link the injections of Salvarsan derivatives with provocation by searching the *Bibliography of Infantile Paralysis* would have been unsuccessful. The difficulty of distinguishing between causal and coincident events must always be a particular problem in medicine. It is known that many of the earliest clinical observations linking smallpox vaccination with poliomyelitis were made directly by polio investigators. These observations were ignored because there was no obvious connection between the two events and because a fuss over these observations would have disturbed the smallpox vaccination program. The number of records which could be retrieved from the literature is therefore smaller than the number of cases observed. As it is very doubtful that

³² Greenberg, et al., "Injections and paralytic poliomyelitis."

³³ See Wyatt, "Provocation poliomyelitis."

³⁴ *Second International Poliomyelitis Conference*, p. 336.

³⁵ L. Rosen and G. Thooris, "Poliomyelitis in French Oceania," *Amer. J. Hyg.*, 1953, 57: 237-52.

³⁶ L. Rosen and G. Thooris, "Aspect clinique d'une épidémie de poliomyélite en océanie française et importance des injections intramusculaires dans l'étiologie des paralysies," *Presse Méd.*, 1952, 60: 1712-14.

³⁷ C. J. Hackett, "Yaws," in E. E. Sabben-Clare, J. Bradley, and K. Kirkwood, eds., *Health in Tropical Africa During the Colonial Period* (Oxford: Oxford University Press, 1980).

³⁸ Hill, "Inoculation procedures," pp. 330-32.

smallpox vaccination does provoke poliomyelitis and it is likely that the association was accidental, the polio investigators were probably right to ignore it.³⁹

The Brodie-Park and Kolmer Vaccines, 1935. An important connection which was apparently missed in the development of knowledge of provocation poliomyelitis was the parallel with two ill-fated experimental vaccines for poliomyelitis used in 1935. Paul wrote that "as far back as 1935 there had been speculations about a possible relationship between the local irritation caused by the Kolmer vaccine and the vaccine-induced paralysis that followed it."⁴⁰ Dr. Tom Rivers and Dr. J. Leake had criticized the vaccines at the meeting of the American Public Health Association in November 1935. Their criticisms were published in late 1935 and early 1936;⁴¹ the publication by Lambert followed later in 1936 in an American journal but if it was read by those in polio research they failed to connect the two events. They probably accepted Lambert's opinion that the injections and paralyzes were coincidental.

As I have pointed out in a previous publication,⁴² there are two kinds of provocation poliomyelitis: that originally described by Martin, Geffen and McCloskey where the injection of an immunogenic or inflammatory substance localizes an existing poliovirus infection, and the provocation poliomyelitis following injection of live poliovirus as in the Cutter Incident of 1955.⁴³ The differences between the two kinds are shown in Table 1. It has always been assumed that the cases of poliomyelitis associated with the Kolmer and Brodie-Park vaccines were caused by virus inoculation as in the Cutter Incident of 1955. In their paper, Nathanson and Langmuir head their section on the *Kolmer and Brodie vaccines* with "Previous occurrence of poliomyelitis due to virus inoculation."⁴⁴ Wilson also assumes that the vaccines contained infectious virions.⁴⁵ However, the Brodie-Park vaccine is supposed to have caused three verified cases of paralysis, occurring twelve, thirteen and fourteen days after the first and only inoculation: the inoculations were made in the summer months and cases of poliomyelitis occurred in unvaccinated control children.⁴⁶ The vaccine cases have an incubation time nearer to that of provocation poliomyelitis by naturally acquired virus than virus-provocation (Table 1). Although Nathanson and Langmuir say that "it is impossible to incriminate or exonerate the Brodie vaccine on epide-

³⁹ Wyatt, "Provocation poliomyelitis."

⁴⁰ Paul, *History of Poliomyelitis*, p. 393.

⁴¹ J. P. Leake, "Poliomyelitis following vaccination against this disease," *JAMA*, 1935, 105: 2152; T. M. Rivers, "Immunity in virus diseases with particular reference to poliomyelitis," *Amer. J. Pub. Health*, 1936, 26: 136-42.

⁴² Wyatt, "Provocation poliomyelitis."

⁴³ N. Nathanson and A. D. Langmuir, "The Cutter Incident: poliomyelitis following formaldehyde-inactivated poliovirus vaccination in the United States during the spring of 1955. II. Relationship of poliomyelitis to Cutter vaccine and III. Comparison of the clinical character of vaccinated and contact cases occurring after use of high rate lots of Cutter vaccine," *Amer. J. Hyg.*, 1963, 78: 29-60, 61-81.

⁴⁴ Nathanson and Langmuir, "Cutter Incident, I," p. 18.

⁴⁵ Wilson, *Hazards of Immunization*, pp. 42-44.

⁴⁶ Nathanson and Langmuir, "Cutter Incident, I," p. 20.

Table 1: Comparison of natural poliomyelitis with poliomyelitis resulting from provocation with and without poliovirus

	Poliomyelitis		
	provocation		natural
	with poliovirus and vaccine	without poliovirus	
Median incubation in days	7 ^a	[12] ^b	12 ^a
Mean number of limbs paralysed 0-4 yr.	2.5 ^c	>1.4 ^b	1.4 ^d
Relative risk	$\times 0.1^e$	$\times 12^f$	1

^a Figure 1, Nathanson and Langmuir, "Cutter Incident, III."

^b *Ibid.*; R. F. Korns, R. M. Albrecht and F. B. Locke, "The association of parenteral injections with poliomyelitis," *Amer. J. Pub. Health*, 1952; 42: 153-69.

^c Table 11, Nathanson and Langmuir, "Cutter Incident, III."

^d Table 12, *Ibid.*

^e From Nathanson and Langmuir, "Cutter Incident, II."

^f Wyatt, "Provocation of poliomyelitis by multiple injections."

miologic grounds," I suggest that it is possible to exonerate it of containing live virus while suggesting that the cases may have resulted from provocation caused by the monkey spinal cord components in the vaccine.⁴⁷ Some inoculations made subcutaneously do not provoke, and in general, only intramuscular injections do. However, the more immunogenic or especially inflammatory the substance, the greater the provocation.⁴⁸ I have suggested that the mechanism of entry of poliovirus to the central nervous system is via motor nerve endings and that this entry is facilitated by lymphocytes. S. Ben-Efraim and D. A. Long showed that the ability of vaccines to provoke poliomyelitis in man was related to their ability to cause skin hypersensitivity in guinea-pigs:⁴⁹ the Brodie-Park vaccine was suspended in monkey spinal cord and Sabin has shown that this component will cause a delayed hypersensitivity reaction in man.⁵⁰ Monkey spinal cord in the Brodie-Park vaccine was given as a ten percent suspension, in a five ml. dose of which one to two ml. were given intradermally and the rest subcutaneously.⁵¹ This could have been enough to involve underlying muscles in the inflammation. Unlike the Kolmer vaccine which occasionally paralyzed test monkeys, the Brodie-Park vaccine failed to affect test monkeys, thus weakening the argument for inherent infectivity. It has been suggested that Brodie took his life following the ill-fated vaccine trials of 1935: it would be ironic if his vaccine had been free of infectivity but provoked poliomyelitis because it was technically not

⁴⁷ *Ibid.*

⁴⁸ Wyatt, "Provocation poliomyelitis."

⁴⁹ S. Ben-Efraim and D. A. Long, "The sensitising power of diphtheria prophylactics in relation to poliomyelitis," *Lancet*, 1957, 2: 1033-35.

⁵⁰ A. B. Sabin, W. H. Park and C. W. Jungeblut, "Nature of skin reactions produced by heat-inactivated poliomyelitis virus," *Arch. Int. Med.*, 1933, 51: 878-89.

⁵¹ Nathanson and Langmuir, "Cutter Incident, I," p. 19.

possible to produce it without monkey spinal cord components. Comment such as "the shoddy, deadly Brodie vaccine" is uncalled for.⁵²

The Kolmer Vaccine. A reassessment of the Kolmer vaccine is much more difficult: were the ten cases of poliomyelitis associated with it caused by virus in the vaccine or by coincident infection? The vaccine was not safe and was rightly withdrawn but what we now know about provocation makes interpretation more difficult than at the time of its use. Was the Kolmer vaccine immunogenic? Kolmer thought so and had given protection to eighteen monkeys vaccinated with his vaccine and challenged by intracerebral inoculation.⁵³ This was repeated by three other groups of workers who failed to protect monkeys challenged by intracerebral,⁵⁴ intranasal,⁵⁵ and intracerebral and intranasal inoculation.⁵⁶ However, the serum of some vaccinated animals did contain neutralizing antibodies.⁵⁷ The vaccine appeared to increase the antibody titre of twenty-one out of twenty-five children. The intracerebral and intranasal inoculation of challenging virus was too severe a test for the presence of serum antibodies, which it is now known will give protection even when present in very low concentration. I therefore consider the Kolmer vaccine to have possessed low immunogenicity.

Did the vaccine contain live virus? Yes, intracerebral inoculation of the virus produced paralysis in all nine test animals and Kolmer appears to have admitted that no appreciable attenuation of the virus occurred in his vaccine.⁵⁸ It is unclear whether Kolmer said this before or after the fateful meeting in St. Louis on 19 November 1935. Kolmer was probably inactivating some of the virions, while leaving a few still virulent, as suggested by Rivers.⁵⁹ The virus was, however, the MV (or VM) strain from the Rockefeller Institute which later work showed would only replicate in neural tissue;⁶⁰ by the subcutaneous route of inoculation used by Kolmer, it was mostly non-infective. Kolmer's vaccine was given on numerous occasions to unprotected monkeys without any monkey contracting poliomyelitis: at least forty-two monkeys were vaccinated by Kolmer⁶¹ and eighteen by Kramer.⁶² However, at least some batches contained live virions.⁶³ Sabin in a se-

⁵² A. E. Klein, *Trial by Fury: The Polio Vaccine Controversy* (New York: Charles Scribner's Sons, 1972), p. 157.

⁵³ J. A. Kolmer, "A successful method for vaccination against acute anterior poliomyelitis: preliminary report," *Amer. J. Med. Sci.*, 1934, 188: 510-14.

⁵⁴ S. D. Kramer and L. H. Grossman, "Active immunization against poliomyelitis: a comparative study. II. Experimental immunization of monkeys with virus treated with ricinoleate," *J. Immunol.*, 1936, 31: 183-89.

⁵⁵ E. W. Schultz and L. P. Gebhardt, "On the problem of immunization against poliomyelitis," *Calif. West. Med.*, 1935, 43: 111-12.

⁵⁶ P. K. Olitzky and H. R. Cox, "Experiments on active immunization against experimental poliomyelitis," *J. Exper. Med.*, 1936, 63: 109-25.

⁵⁷ Kramer and Grossman, "Active immunization"; Schultz and Gebhardt, "Problem of immunization."

⁵⁸ Kramer and Grossman, "Active immunization." See footnote on p. 189.

⁵⁹ Rivers, "Immunity in virus diseases."

⁶⁰ Paul, *History of Poliomyelitis*.

⁶¹ J. A. Kolmer, "Susceptibility and immunity in relation to vaccination in acute anterior poliomyelitis," *JAMA*, 1935, 105: 1956-62; *idem*, "An improved method of preparing the Kolmer poliomyelitis vaccine," *Amer. J. Pub. Health*, 1936, 26: 149-57.

⁶² Kramer and Grossman, "Active immunization."

⁶³ Nathanson and Langmuir, "Cutter Incident, I," p. 18.

ries of experiments found that no cases of paralysis followed intracutaneous or subcutaneous injection of the MV strain in the abdominal region (twelve monkeys) or subcutaneously into the tonsils (six monkeys).⁶⁴ In spite of the low infectivity resulting from subcutaneous inoculation of the MV strain (a few cases of paralysis in monkeys),⁶⁵ the nine or ten cases out of about 12,000 receiving vaccine could well have resulted from residual virus in the vaccine.

More damning is the correlation of initial paralysis with site of inoculation, the high fatality rate, and the incubation time, all being associated with provocation. None of these necessarily implies provocation linked to virus contained in the vaccine. The vaccine contained three components which could have provoked: four percent monkey spinal cord, one percent sodium ricinoleate and 1: 80,000 phenyl-mercuri-nitrate and in addition, some batches were contaminated with bacteria.⁶⁶ "The injections were accompanied by some stinging pain"⁶⁷ and by "local reactions comparable to those produced by diphtheria toxoid," in about nine percent of vaccines.⁶⁸ Among eighty-five nurses given the vaccine, five had severe constitutional reactions.⁶⁹ Without containing any virus, these components alone would probably provoke.

In the Cutter Incident of an outbreak of poliomyelitis following vaccination in 1955, the correlation of initial paralysis with the site of injection was eighty-two percent;⁷⁰ correlation in the Kolmer cases was four out of nine with the possibility of two other cases where the site of injection is not known.⁷¹ This is strange because the MV strain only replicates in neural tissue and does not produce a viremia; one would expect one hundred percent correlation if provocation resulted from virus contained in the vaccine. The incubation periods of the Kolmer cases would fit equally with provocation with injected virus (Cutter vaccinees⁷²) or provocation with coincident virus infection (natural poliomyelitis⁷³). The high case fatality rate might have resulted from the monkey spinal cord component, especially as all the fatal cases received two doses of vaccine, the second acting as a booster as previously described.⁷⁴ Again this would apply equally to injected or wild virus infection.

Leake drew attention to the lack of epidemic cases in any of the areas

⁶⁴ A. B. Sabin, "Experimental poliomyelitis by the tonsillopharyngeal route with special reference to the influence of tonsillectomy on the development of bulbar poliomyelitis," *JAMA*, 1938, 111: 605-10.

⁶⁵ *Ibid.*; Nathanson and Langmuir, "Cutter Incident, I," p. 18.

⁶⁶ Kolmer, "Improved method," J. A. Kolmer, "Vaccination against acute anterior poliomyelitis," *Amer. J. Pub. Health*, 1936, 26: 126-35.

⁶⁷ Kolmer, "Successful method."

⁶⁸ Kolmer, "Vaccination."

⁶⁹ J. F. Kessel, "Discussion of poliomyelitis papers," *Amer. J. Pub. Health*, 1936, 26: 145-47.

⁷⁰ Nathanson and Langmuir, "Cutter Incident, II."

⁷¹ I have not included Kolmer's case no. 1, in Kolmer, "Vaccination" which is not listed by Leake, "Poliomyelitis following vaccination."

⁷² Nathanson and Langmuir, "Cutter Incident, I."

⁷³ P. E. Sartwell, "The incubation period of poliomyelitis," *Amer. J. Pub. Health*, 1952, 42: 1403-8.

⁷⁴ Wyatt, "Poliomyelitis an auto-allergic disease."

where Kolmer cases occurred. Using the highest incidence of reported cases, he judged that the chance of a case occurring should have been one in ten, "yet a case occurred. The likelihood of the whole series of cases having occurred through natural causes is extremely small."⁷⁵ Nathanson and Langmuir quote this and appear to agree with the conclusions.⁷⁶ Leake's reasoning is faulty because it is now known that provocation can increase susceptibility to paralysis. In the case of a single injection such as DPT, it may be increased by up to $\times 5$, and single or multiple injections of Salvarsan etc. may increase the paralytic rate by $\times 25$.⁷⁷ Thus the Kolmer cases could have been due to provocation with coincident wild virus infection, increasing the susceptibility of the vaccinees by up to twenty-five fold and the lack of non-vaccinee cases would be explained. In two areas of New Jersey wild virus was circulating during the time that Kolmer cases occurred.⁷⁸

At this stage it might be wise to say that the Kolmer vaccine was probably a potent provoking agent. Although the cases might have been due to virus from the vaccine or in some cases at least might have been caused by concurrent wild virus infection, the vaccine was probably ineffective at giving immunity even to Type 2 and that, for whatever reason, the vaccine was very unsafe (Table 2).

Tonsillectomy. From the beginning, provocation poliomyelitis was linked by Martin to the increased susceptibility following tonsillectomy.⁷⁹ This linking must have been due at least partly to the similar incubation periods of seven to eighteen days, although this is also the incubation period for natural poliomyelitis. A convincing explanation of the effect of tonsillectomy is still awaited.

John R. Paul and A History of Poliomyelitis. John R. Paul has written a magnificent history of poliomyelitis and it therefore seems carping to criticize such an excellent book, yet his treatment of provocation furnishes, I believe, a clue to poliomyelitis research.⁸⁰ Provocation poliomyelitis following DPT inoculations appears in Paul's book only in discussion of the trials of prophylactic gammaglobulin in 1951 where it is referred to as "The matter of trauma inflicted by injections was also brought up. This was indeed a theoretical menace" (p. 393). The whole of the discussion on provocation poliomyelitis occupies nine lines: the words used in this section: England, Australia; diphtheria, whooping cough and tetanus; trauma and irritant; do not appear in the index and the chapter is headed "The bloodstream as a pathway in the human body." There are no references to provocation in the

⁷⁵ Leake, "Poliomyelitis following vaccination."

⁷⁶ Nathanson and Langmuir, "Cutter Incident, I," p. 18.

⁷⁷ H. V. Wyatt, "Provocation of poliomyelitis by multiple injections: experience with congenital syphilis," in *Reports of VIIth International Congress of Infections and Parasitic Diseases* (Varna, Bulgaria, 1978), 1: 420-23.

⁷⁸ H. F. Vaughan, "Discussion of poliomyelitis papers," *Amer. J. Pub. Health*, 1936, 26: 143-44.

⁷⁹ Martin, "Local paralysis,"; Aycock and Luther, "Occurrence of poliomyelitis."

⁸⁰ Paul, *History of Poliomyelitis*.

Table 2. Comparison of the three vaccines associated with provocation poliomyelitis.

Vaccine	Brodie-Park, 1935	Kolmer, 1935	Cutter, 1955
Virus strain	MV	MV	Mahoney
properties	no viremia	no viremia	viremia
infectivity with subcutaneous inoculation	very low	very low	normal
injection	intradermal and subcutaneous	subcutaneous	intramuscular
vaccine	formalin-inactivated 5% monkey spinal cord	live, attenuated(?) 2% monkey spinal cord 1% sodium ricinoleate 1:80 000 phenyl-mercuri-nitrate	formalin-inactivated tissue culture debris
live virions present	probably none	very small number	small number
number of cases of paralysis per 1000 vaccinees	0.3	1	1
provocation	provocation with wild virus	some provocation with wild virus some provocation with vaccine virus	provocation with vaccine virus

footnotes. Provocation does not appear in the index and only once in the text,—“gamma-globulin injections were not irritating and could safely be given without fear of a provocative action.” In this section, Paul says that “as far back as 1935 there had been speculations about a possible relationship between the local irritation. . . . and vaccine-induced paralysis” but this is not mentioned in the chapter on the “ill-fated vaccine trials of 1935.” This is in marked contrast to Paul’s discussion of tonsillectomy (pp. 187–89) and the Cutter Incident (pp. 435–37.) However, in Paul’s index “Cutter incident” refers the reader to pp. 260, 408, 438, 450, 461, 466, 467 on each of which there is a reference to the incident but the crucial entry of pages 435–37 is omitted.

I believe that there has been a continuous wish to confront poliomyelitis as a major disease—but not to make it too major or the problem would have been overwhelming. There has often been a reluctance to face epidemics where there has been a high rate of paralysis, e.g., Kern’s twenty-four percent, Lambert’s seven point eight percent, or the implications of an increase in paralysis of $\times 12$ by provocation. This is a subject needing further investigation.

It is interesting to speculate on how things would have differed if provocation poliomyelitis had been recognized earlier. If Lambert had realized that the poliomyelitis epidemic was caused by the injections for yaws, he might have published in a different journal and searched for other papers describing poliomyelitis after injections with Salvarsan derivatives. The

recognition of provocation in 1935 might have provided a key to the pathogenesis of poliomyelitis: it certainly would have focussed attention on the route of invasion of the central nervous system (CNS) and the possibility of an early viremia. The very fact that the first signs of CNS involvement occurred about eight days after the provoking injection might have stimulated research into this phase of infection and experiments with provocation in animals would then have been carried out. It is fairly certain that a provoking injection which did not contain poliovirus, combined with CNS injection or intranasal infection of poliovirus, would not have resulted in provocation poliomyelitis and so might have led to a search for another route of infection more in accord with natural and provocation poliomyelitis. It is not clear whether the MV strain then used at the Rockefeller Institute, when injected with, e.g., Salvarsan into a muscle of a monkey would have caused paralysis, because it was a neurotropic strain only. The inflammation caused by the Salvarsan and by the ten percent monkey spinal cord in which the virus was suspended, would probably have enabled the MV strain to enter the motor nerves and hence paralysis would have resulted.⁸¹ The MV strain did not cause a viremia, so the viremic stage would have been missed. However, the experiment might have led others to see if injection of fresh material from a patient plus Salvarsan into a muscle, would cause paralysis. This might have led to the finding of viremia and attempts to discover what was happening in the days before signs of CNS involvement. The weight of opinion in 1936 was firmly based on the high neurotropism of poliovirus (the MV strain which was passaged in monkeys) and it was not until 1938 that Trask and Paul showed that newly isolated strains were not so neurotropic.⁸² They showed that these strains produced paralysis when introduced into the skin of monkeys, unlike the MV strain.

However, it is more likely that even if Lambert had made the connection, nothing would have come of it until after the Second World War. As Paul records about the ill-fated vaccine trials, "The events of 1935 cut more deeply. . . . so deeply and traumatically (sic!)." Lambert's paper came soon after the trauma of the vaccine trials: perhaps if it had been published earlier or later it would have had more impact. Albert Sabin in this period was laying the foundations for future progress by a series of fundamental studies: he would have explored provocation.⁸²

Conclusion. Sound clinical observation is still the basis of much initial medical insight. However, the individual clinician is seldom able to publish details of more than a few cases and the reports, which may therefore appear anecdotal, are likely to be unnoticed as this study of provocation poliomyelitis shows, unless there is a theoretical basis for further investigation. This should be kept in mind by medical historians. Further, clinical

⁸¹ Wyatt, "Provocation poliomyelitis."

⁸² J. D. Trask and J. R. Paul, "The skin infectivity of poliomyelitis virus," *Science*, 1938, 87: 44-45.

observation is by no means confined to doctors, as the experience in Samoa shows.⁸³ After the papers on provocation appeared in the *British Medical Journal* in 1950, Price wrote a letter from the Belgian Congo:

The paper by Professor A. Bradford Hill and Dr. J. Knowelden (July 1, p. 1) and your annotation (p. 28) throw light on what is known here among the African natives as "injection paralysis." This condition is seen from time to time in the out-patient department of this hospital and the story is very similar in each case. The baby had a fever; he was taken to a dispensary, where an injection of quinine was given into a buttock, "since when the baby's leg has been paralysed." It always seems to be the leg which was injected. However, careful questioning of the mother usually makes it clear that the paralysis did not appear until two or three weeks after the injection, and there has never been any evidence of suppuration of the injection site. Clinically, the condition is poliomyelitis; the condition is seen among older children who have not had an injection. The leg is flaccid and there are no sensory disturbances. Most native dressers are taught carefully how to avoid injections into the region of the sciatic nerve.⁸⁴

It appears, then, that "injection paralysis" has scientific grounds for its existence, and that indeed the localization in the affected limb is a direct effect of the injection, as Congo mothers have so long asserted to sceptical doctors.

The compilation of bibliographies therefore does not necessarily help the reader to find these isolated reports: they are likely to be indexed and abstracted according to the prevailing ideas.⁸⁵ However, this is not to deny that bibliographies can still be extremely useful when further search is required after a theoretical framework has been suggested.

⁸³ Lambert, "Yaws campaign."

⁸⁴ E. W. Price, "Inoculation and injection paralysis," *Brit. Med. J.* 1950, 2: 1391.

⁸⁵ For an additional example, in 1930, two doctors in Madagascar published a short note on paralysis following injections of quinine given to small children. The parents had first (in 1924) attributed paralysis to the injections and so had other doctors. The authors distinguished between paralysis following these injections and the trauma following badly placed injections. They followed this with an even shorter note in 1933; neither paper was recorded in the *Bibliography*. See Fontoynt and Raharjaona, "La poliomyélite antérieure aiguë (paralysie infantile) à Madagascar," *Bull. Soc. Path. Exot.*, 1930, 23: 554-55; 1933, 26: 1118-19.